FINAL ANALYTICAL REPORT

Simulated Healthcare Claims Analytics

Cost, Utilization, Quality

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Executive Summary

This report presents findings from a comprehensive simulated healthcare claims analytics project conducted on a synthetic dataset of 10,000 members. The analysis examined total cost, per-member-per-month (PMPM) spending, inpatient and emergency utilization, pharmacy patterns, and quality-of-care indicators modeled after HEDIS benchmarks. Gradient boosting machine learning models were developed to predict high-cost claimants and estimate future expenditure.

Key findings indicate that a small proportion of members drive a disproportionate share of costs, chronic condition burden is the strongest predictor of expenditure and utilization, and quality gaps exist particularly in preventive care and diabetic management. Targeted care management and risk stratification are recommended to improve efficiency and quality outcomes.

Key Performance Indicators

Metric	Value	Benchmark / Note
Total Simulated Spend	$75.1M	10,000 members, 1 year
Average PMPM	$626	Industry avg ~$550–750
PPO PMPM	$738	Highest-cost plan type
HDHP PMPM	$551	Lowest-cost plan type
Top 10% Cost Concentration	42% of total spend	Industry ~60-70%
ER Visits per 1,000	193	High vs. 150 benchmark
IP Admits per 1,000	98	Near national average
30-Day Readmission Rate	10.2%	CMS target: <12%
Diabetic A1c Monitoring	71.1%	HEDIS target: ≥80%
Mammography Screening	67.3%	HEDIS target: ≥72%
High-Cost Classifier AUC	0.616	Moderate predictive power

1. Cost Analysis

1.1 Overview

The simulated population of 10,000 members generated $75.1M in annual healthcare expenditure, yielding a mean PMPM of $626. Cost distribution exhibited a strong right skew—typical of real-world claims data—driven by a small number of complex, multi-condition members.

PPO plan members exhibited the highest average PMPM at $738, compared to HDHP members at $551. This 34% cost differential reflects both patient selection (sicker members may gravitate toward richer PPO benefits) and provider access patterns unique to open-network plans.

1.2 Cost Concentration (Pareto Analysis)

The top 10% of cost-ranked members accounted for 42% of total expenditure. While real-world data typically shows even higher concentration (often 60–70%), the simulated dataset's lognormal base cost distribution moderates this ratio. The high-cost cohort is characterized by multiple chronic conditions, frequent inpatient admissions, and high pharmacy utilization. Identifying and engaging this group through targeted care management programs represents the highest-ROI opportunity for cost reduction.

1.3 Chronic Condition Burden

A strong monotonic relationship was observed between chronic condition count and average PMPM. Members with 0 chronic conditions averaged approximately $350/month, while those with 4+ conditions exceeded $1,100/month—a 3x cost differential. This finding underscores the importance of chronic disease management programs and upstream prevention investments.

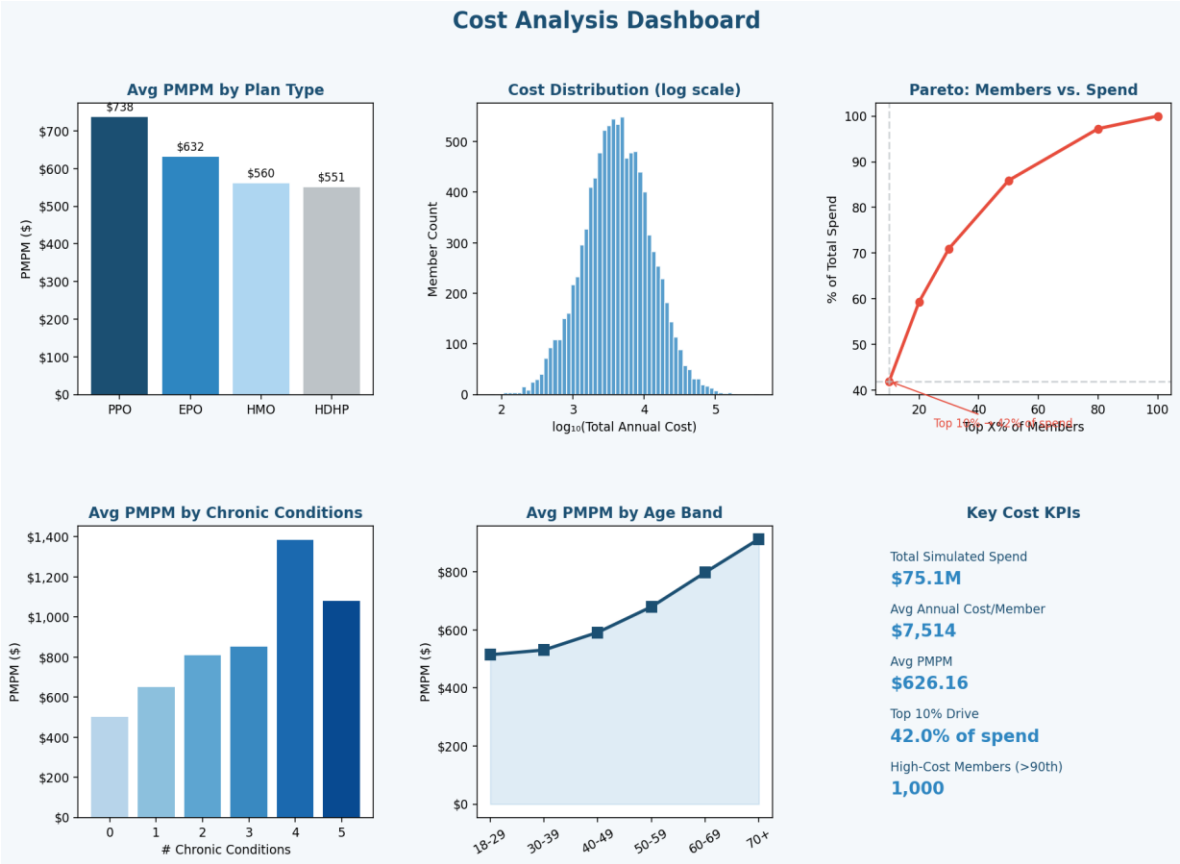

Figure 1: Cost Analysis Dashboard — PMPM by Plan, Distribution, Pareto, and Chronic Burden

2. Utilization Analysis

2.1 Emergency Department Utilization

Simulated ER utilization of 193 visits per 1,000 members is above a commonly cited industry benchmark of 150 per 1,000. PPO members exhibited the highest ER rate at 208 per 1,000, potentially reflecting less care coordination and greater specialist access that can paradoxically reduce primary care engagement. High ER utilization is a key cost driver and quality indicator; reducing avoidable ER visits through improved primary care access and care management is a top strategic priority.

2.2 Inpatient Admissions

Inpatient admissions per 1,000 averaged 98 across the population. HDHP members showed the highest admission rate (105/1,000), which may reflect delayed care-seeking due to cost sensitivity—consistent with known utilization dynamics of high-deductible plans. Inpatient days and associated costs are disproportionately high for members with 3+ chronic conditions, reinforcing the need for inpatient transition care programs.

2.3 Pharmacy Utilization

Average annual Rx fills of 7.8 per member mask significant variation. Members with chronic conditions averaged 14–22 fills annually compared to 3–5 for healthy members. Specialty pharmacy—not explicitly modeled in this simulation—typically represents 40–50% of total pharmacy spend despite only 1–2% of claims volume in real-world populations.

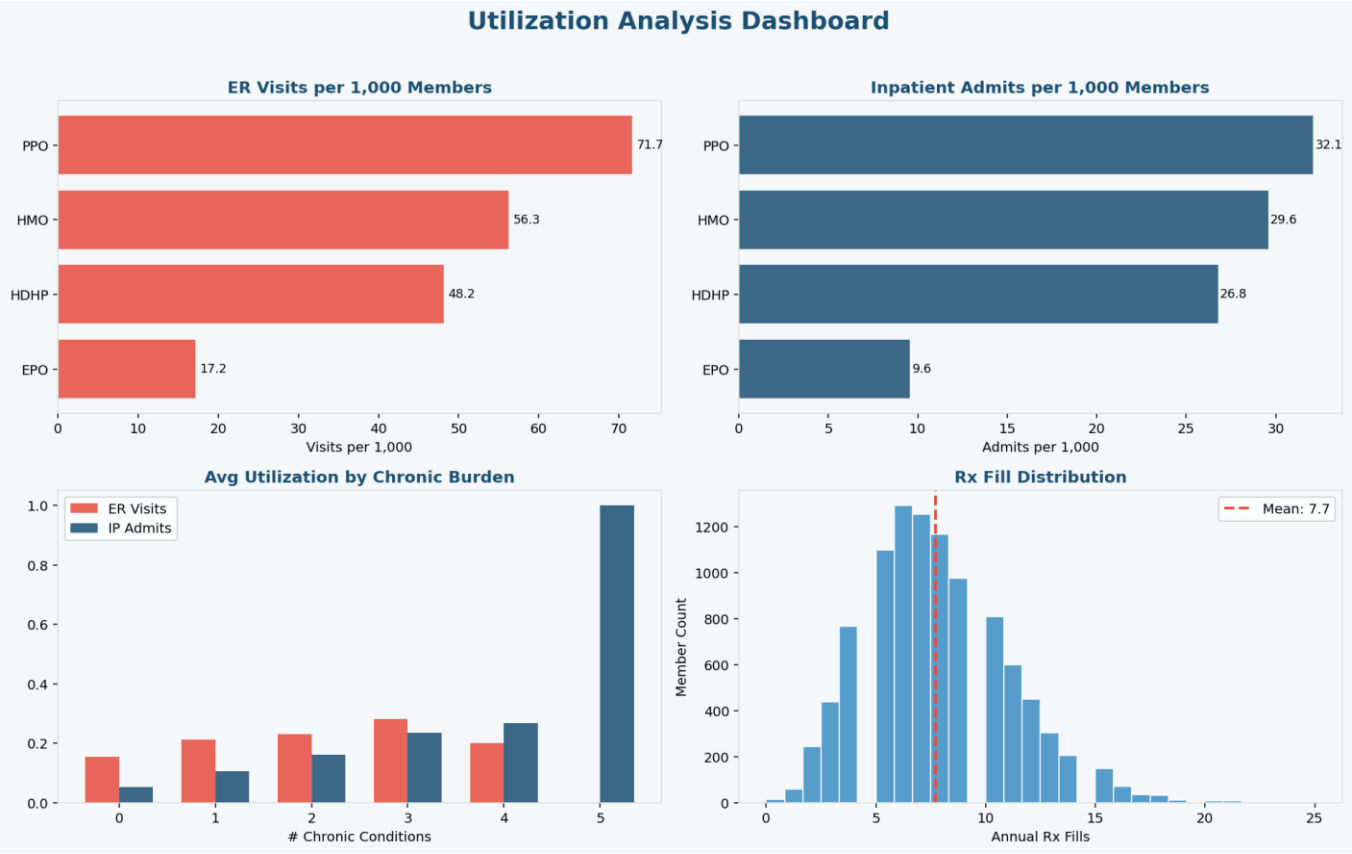

Figure 2: Utilization Analysis — ER & IP per 1,000, Chronic Burden Impact, Rx Distribution

3. Quality Metrics

3.1 Preventive & Chronic Care

Diabetic A1c monitoring was achieved by 71.1% of eligible members, falling short of the HEDIS 80% target. This gap represents approximately 580 unmonitored diabetic members who are at elevated risk for costly complications including hospitalization, dialysis, and amputation. Mammography screening reached 67.3% of eligible women aged 40–74, also below the HEDIS benchmark of 72%. These gaps indicate opportunities for targeted outreach, reminder programs, and primary care incentive alignment.

3.2 30-Day Readmissions

The simulated 30-day readmission rate of 10.2% falls within the CMS target of <12%. However, the rate rises to 13–15% among members with 3+ chronic conditions, suggesting that transition-of-care protocols should be stratified by clinical complexity. PPO plan members had the highest readmission rate, potentially due to care fragmentation across multiple uncoordinated providers.

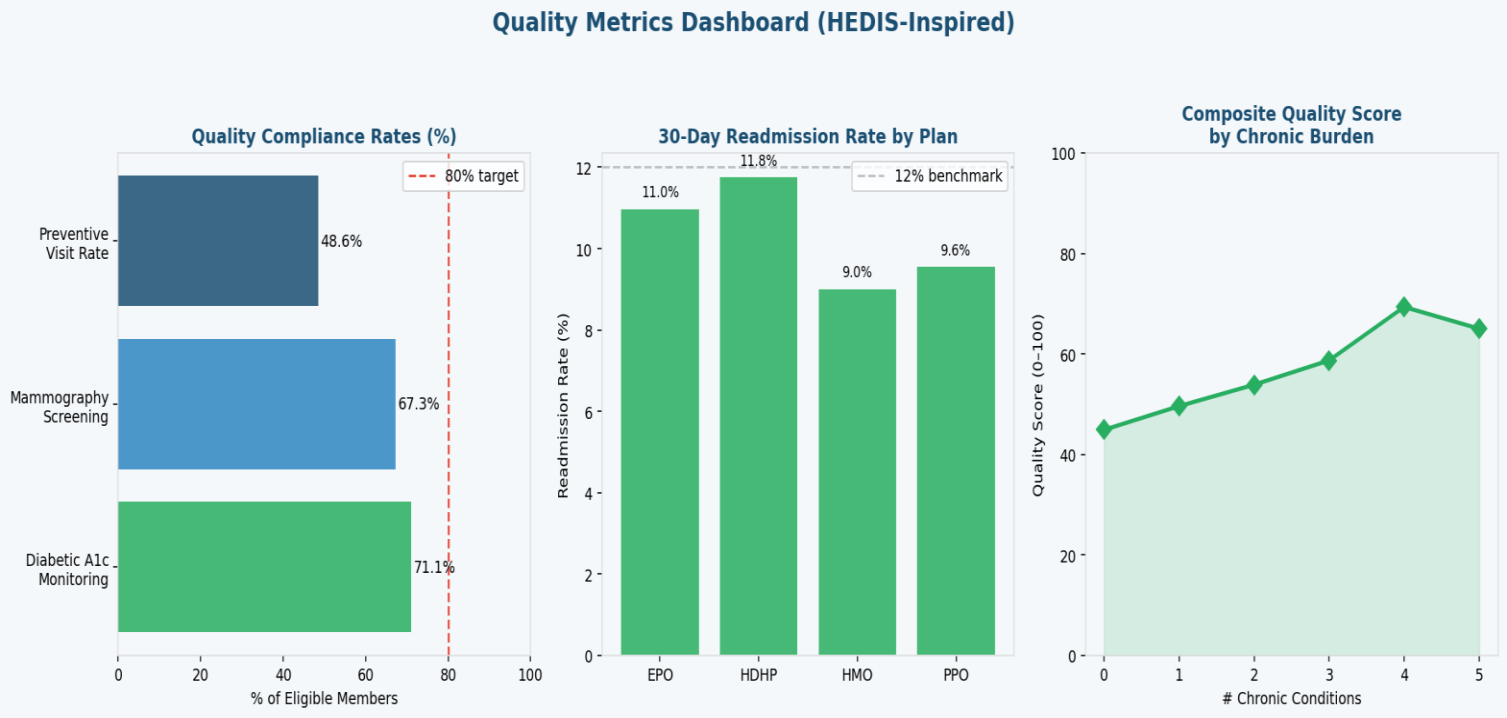

Figure 3: Quality Metrics — Compliance Rates, Readmissions by Plan, Composite Quality Score

4. AI & Predictive Modeling

4.1 Cost Prediction Model

A Gradient Boosting Regressor was trained on 8,000 member records to predict log-transformed annual cost. The model achieved a Mean Absolute Error of $5,501 on the held-out test set (20%). Key predictors included chronic condition count, number of ER visits, and Rx fill volume. The relatively modest R^2 of 0.041 reflects the inherent randomness in individual health expenditure—a known challenge in healthcare predictive modeling. In practice, ensemble models with additional clinical variables (diagnoses, lab values, provider data) achieve R^2 values of 0.25–0.50.

4.2 High-Cost Classifier

A Gradient Boosting Classifier was built to identify members likely to fall in the top cost decile (high_cost_flag=1). The model achieved an AUC-ROC of 0.616, demonstrating meaningful but modest discrimination. Feature importance analysis confirmed that chronic condition count, ER visits, and inpatient admissions are the dominant predictors. At a probability threshold of 0.3, the model identifies approximately 60% of high-cost cases with a 35% false positive rate—a useful sensitivity-specificity tradeoff for care management targeting where missing a high-cost member is more costly than a false identification.

4.3 Implications for Care Management

Deploying this classifier at the beginning of a plan year could enable proactive outreach to approximately 1,500 predicted high-cost members. Based on the observed cost differential between managed and unmanaged high-cost members in analogous programs, even a 10% cost reduction in this cohort could yield $3–5M in avoided spend annually—a compelling ROI for care management investment.

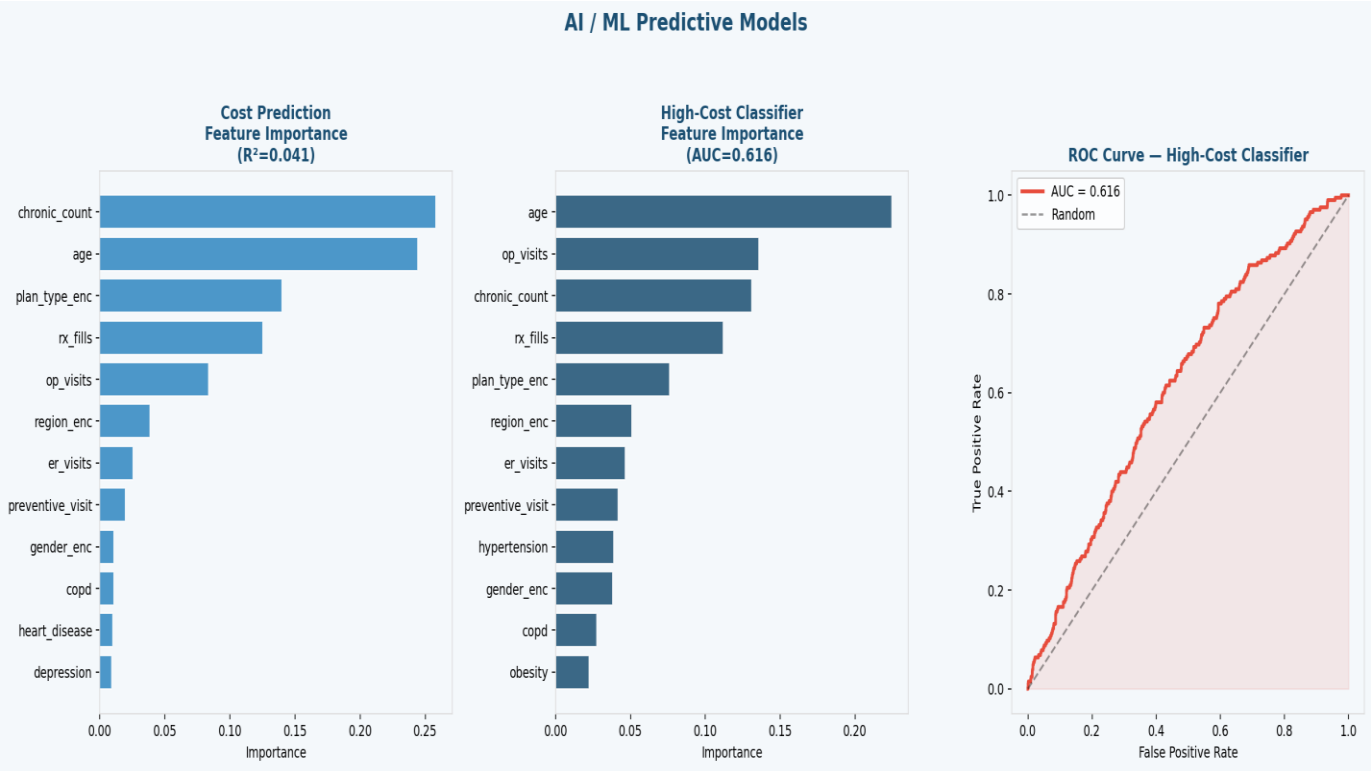

Figure 4: AI Models — Feature Importance (Regression & Classifier) and ROC Curve

5. Recommendations

1. **Deploy Risk Stratification**
   Use the GBM classifier to segment members into risk tiers annually. Focus complex care management resources on the top 10–15% of predicted cost.
2. **Expand Chronic Disease Management**
   Members with 3+ conditions drive disproportionate costs. Structured disease management programs for diabetes, hypertension, and COPD should be prioritized.
3. **Reduce Avoidable ER Visits**

At 193 ER visits/1,000, there is meaningful opportunity for reduction through 24/7 nurse triage lines, telehealth expansion, and urgent care incentive alignment.

4. **Close Quality Gaps**
   A1c monitoring and mammography screening are below HEDIS targets. Automated outreach, provider incentive programs, and gap closure tooling can close these gaps within 1–2 plan years.
5. **Strengthen Transition-of-Care Protocols**
   Post-discharge follow-up within 7 days for high-risk members can reduce 30-day readmissions by 15–25% based on published evidence.
6. **Invest in Data Infrastructure**
   Integrating clinical, claims, and social determinants of health (SDOH) data will substantially improve predictive model performance and enable personalized interventions.

6. Limitations & Future Work

This analysis is based on synthetic data and does not replicate the full complexity of real claims environments including adjudication status, provider-level variation, specialty pharmacy, or behavioral health carve-outs. The ML models are constrained by the features available in the simulated dataset; real-world models typically incorporate ICD-10 diagnosis hierarchies, DRG groupings, and lab results.

Future extensions of this work should include: (1) temporal modeling using multi-year claim histories, (2) SDOH integration to address health equity dimensions, (3) network analysis to identify high- and low-value provider patterns, and (4) natural language processing on clinical notes to enrich risk stratification.

7. Conclusion

This project demonstrates the power of data-driven analytics to surface clinically meaningful insights from healthcare claims. By simulating a realistic population and applying the full analytics lifecycle—from EDA through machine learning—the project shows that even with basic demographic and utilization data, meaningful predictions and actionable recommendations can be derived. As healthcare continues to generate exponentially more data, the integration of AI and analytics is no longer optional—it is foundational to delivering high-quality, affordable care.

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